

JOSHUA CARBON

AI DEVELOPER / COMPUTER VISION

Computer Vision • AI Assistants • Prompt Engineering

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SUMMARY

Computer Engineering graduate focused on practical AI application: a custom-trained YOLOv5 computer-vision system for Japanese Kanji recognition (95.33% accuracy, 50 classes, 150 test trials), and an AI assistant (CompetencyIQ) built in Microsoft Copilot Studio using prompt engineering for workforce competency intelligence. Also delivered AI-assisted functionality inside a real client software platform (AIQUANTA), connecting AI work to production software integration rather than isolated experimentation.

SKILLS

AI & Computer Vision: YOLOv5, OpenCV, OCR, Image Processing, Roboflow, Google Colab

AI Tools: Microsoft Copilot Studio, Prompt Engineering, AI Agents/Assistants

Programming: Python, C++, C, C#

Web Development: React, TypeScript, Vite, React Router, HTML, CSS

Software Development: OOP, Windows Forms, Git, Debugging

Embedded / IoT: Raspberry Pi, Arduino, ESP32/ESP8266, Sensor Integration, PCB Design

PROJECTS

AI-Based Japanese Character Recognition using YOLOv5 — Thesis project. Prepared and managed training data with Roboflow, trained a custom YOLOv5 object-detection model in Google Colab, and built a Python/OpenCV OCR pipeline deployed on Raspberry Pi for real-time Kanji recognition. Evaluated the model at 95.33% accuracy across 50 Kanji classes and 150 test trials.

CompetencyIQ AI — Built an AI assistant using Microsoft Copilot Studio and prompt engineering to support workforce competency intelligence, structuring its behavior around employee profiles, training history, post-training assessments, technical certifications and expiry, training-needs analysis, audit/quality findings, and operational risk indicators.

QA Automation Framework — Built a reusable web automation framework using Python, Robot Framework, Selenium, and SeleniumLibrary, with reusable automation components, automated test execution, and reporting.

Personal Portfolio Website — Designed and built a React, TypeScript, and Vite portfolio site with React Router-based project case-study pages, a role-tailored resume selector, responsive/adaptive layouts, and technical visualizations; deployed to Netlify with iterative debugging and refinement.

AI-ENABLED CLIENT SOFTWARE

AIQUANTA Training Consultancy Services — Paid freelance/client project. Implemented a client-facing consulting website and operations platform using React, TypeScript, and Vite, translating client requirements into a responsive, component-based frontend. Integrated a Cloudflare Worker HTTP API with D1 relational storage for data handling, built authenticated administration and reporting/analytics features, and added AI-assisted inquiry-relevance functionality alongside Microsoft Graph email integration. Handled deployment, technical SEO, and security considerations, refining functionality through iterative client-driven development and debugging.

EXPERIENCE

Intern / OJT — Erovoutika Robotics and Automation Solution | May 2023 – Aug 2023

- Worked alongside engineering teams to troubleshoot technical issues across automation systems and embedded components.
- Performed debugging and calibration on automation units, supporting consistent performance during operation.
- Supported integration of sensors and embedded components into robotics and automation systems, including research/development work on a line-following robot.
- Authored technical documentation covering setup, calibration, and troubleshooting procedures for engineering reference.

EDUCATION

Bachelor of Science, Computer Engineering — Mapúa University | 2019 – 2025

Bachelor of Science, Electrical Engineering — Mapúa University | 2014 – 2019